KAIZEN MATHS

# Lesson starter

**Algebraic Simplification**

Advanced Simplification

Name _____

**S1**

Simplify:

$-4x\,y^2 + 2x\,y^2 + 4x^2\,y - 5x^2\,y$

**S2**

Simplify:

$-2x^2\,y - 4x^2\,y - 2x\,y^2 - 4x\,y^2 - 3x^3\,y^2 - 2x^3\,y^2 + 8$

**S3**

Simplify:

$4a^2 - 3a^2 + b^2 - 5b^2 - 4a\,b + 5a\,b$

**S4**

Simplify:

$2m(7m + 5) + 3m(7m - 7)$



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**S4**

Simplify:

$2m(7m + 5) + 3m(7m - 7)$

ALG-3B24-L3 · S

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# Exit ticket

**Algebraic Simplification**

Advanced Simplification

Name _____

**E1**

Simplify:

$5p \times 6q^3 \times 4 \times 5p^2$

**E2**

Simplify:

$-2x\,y^2 - 5x\,y^2 + 6x\,y - 2x\,y$

**E3**

Simplify:

$3x^2\,y + 3x^2\,y + 4x\,y^2 + 3x\,y^2 - 3x^3\,y^2 + 2x^3\,y^2 + 5$



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Exit ticket

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Name _____

E1

Simplify:

$5p \times 6q^3 \times 4 \times 5p^2$

E2

Simplify:

$-2x\,y^2 - 5x\,y^2 + 6x\,y - 2x\,y$

E3

Simplify:

$3x^2\,y + 3x^2\,y + 4x\,y^2 + 3x\,y^2 - 3x^3\,y^2 + 2x^3\,y^2 + 5$

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E1

Simplify:

$5p \times 6q^3 \times 4 \times 5p^2$

E2

Simplify:

$-2x\,y^2 - 5x\,y^2 + 6x\,y - 2x\,y$

E3

Simplify:

$3x^2\,y + 3x^2\,y + 4x\,y^2 + 3x\,y^2 - 3x^3\,y^2 + 2x^3\,y^2 + 5$

ALG-3B24-L3 · E

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Exit ticket

Algebraic Simplification

Advanced Simplification

Name _____

E1

Simplify:

$5p \times 6q^3 \times 4 \times 5p^2$

E2

Simplify:

$-2x\,y^2 - 5x\,y^2 + 6x\,y - 2x\,y$

E3

Simplify:

$3x^2\,y + 3x^2\,y + 4x\,y^2 + 3x\,y^2 - 3x^3\,y^2 + 2x^3\,y^2 + 5$

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# Teacher answers & working

**Algebraic Simplification · Advanced Simplification**

ALG-3B24-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## S1

Simplify:
$-4x\,y^2 + 2x\,y^2 + 4x^2\,y - 5x^2\,y$

**Answer:**

$-2x\,y^2 - x^2\,y$

**Working:**

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-4x\,y^2 + 2x\,y^2 + 4x^2\,y - 5x^2\,y \;=\; (-4+2)x\,y^2 \;+\; (4+-5)x^2\,y$

$=\; -2x\,y^2 - x^2\,y$

Step 2: Combined like products with identical variable patterns.

Step 3: Write each simplified group once in the final expression.

## S2

Simplify:
$-2x^2\,y - 4x^2\,y - 2x\,y^2 - 4x\,y^2 - 3x^3\,y^2 - 2x^3\,y^2 + 8$

**Answer:**

$-6x^2\,y - 6x\,y^2 - 5x^3\,y^2 + 8$

**Working:**

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-2x^2\,y - 4x^2\,y - 2x\,y^2 - 4x\,y^2 - 3x^3\,y^2 - 2x^3\,y^2 + 8 \;=\; (-2+-4)x^2\,y \;+\; (-2+-4)x\,y^2 \;+\; (-3+-2)x^3\,y^2 \;+\; 8$

$=\; -6x^2\,y - 6x\,y^2 - 5x^3\,y^2 + 8$

Step 2: Grouped by identical variable products.

Step 3: Write each simplified group once in the final expression.

## S3

Simplify:
$4a^2 - 3a^2 + b^2 - 5b^2 - 4a\,b + 5a\,b$

**Answer:**

$a^2 - 4b^2 + a\,b$

**Working:**

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$4a^2 - 3a^2 + b^2 - 5b^2 - 4a\,b + 5a\,b \;=\; (4+-3)a^2 \;+\; (1+-5)b^2 \;+\; (-4+5)a\,b$

$=\; a^2 - 4b^2 + a\,b$

Step 2: Each group shares the same variable pattern.

Step 3: Write each simplified group once in the final expression.

# Teacher answers & working

**Algebraic Simplification · Advanced Simplification**

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## S4

Simplify:
$2m(7m+5)+3m(7m-7)$

**Answer:**

$35m^2-11m$

**Working:**

Step 1: Expand every bracket first by multiplying each term inside the bracket.

$2m(7m+5)+3m(7m-7)$

$= 14m^2+10m+21m^2-21m$

$= (14+21)m^2 \; + \; (10-21)m$

$= 35m^2-11m$

Step 2: Multiplying $m$ by $m$ gives $m^2$; multiplying $m$ by a constant gives a $m$ term.

Step 3: Collect the like terms and write the simplified expression.

## E1

Simplify:
$5p \times 6q^3 \times 4 \times 5p^2$

**Answer:**

$600p^3\,q^3$

**Working:**

Step 1: Separate the number factors from the variable factors.

$5p \times 6q^3 \times 4 \times 5p^2$

$= (5 \times 6 \times 4 \times 5) \times (p \times p^2) \times q^3$

$= 600 \times p^3 \times q^3$

$= 600p^3\,q^3$

Step 2: The same variable is multiplied, so its powers are added: $p \times p^2 = p^3$.

Step 3: Multiply the coefficients and add powers of the same variable.

# Teacher answers & working

**Algebraic Simplification · Advanced Simplification**

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## E2

Simplify:
$-2x\,y^2 - 5x\,y^2 + 6x\,y - 2x\,y$

**Answer:**

$-7x\,y^2 + 4x\,y$

**Working:**

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-2x\,y^2 - 5x\,y^2 + 6x\,y - 2x\,y \;=\; (-2 + -5)x\,y^2 \;+\; (6 + -2)x\,y$

$=\; -7x\,y^2 + 4x\,y$

Step 2: Combined like products with identical variable patterns.

Step 3: Write each simplified group once in the final expression.

## E3

Simplify:
$3x^2\,y + 3x^2\,y + 4x\,y^2 + 3x\,y^2 - 3x^3\,y^2 + 2x^3\,y^2 + 5$

**Answer:**

$6x^2\,y + 7x\,y^2 - x^3\,y^2 + 5$

**Working:**

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$3x^2\,y + 3x^2\,y + 4x\,y^2 + 3x\,y^2 - 3x^3\,y^2 + 2x^3\,y^2 + 5 \;=\; (3 + 3)x^2\,y \;+\; (4 + 3)x\,y^2 \;+\; (-3 + 2)x^3\,y^2 \;+\; 5$

$=\; 6x^2\,y + 7x\,y^2 - x^3\,y^2 + 5$

Step 2: Grouped by identical variable products.

Step 3: Write each simplified group once in the final expression.